Nonelementary Reaction Kinetics-II

CENG252

Chemical Reaction Engineering

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Pseudo Steady-State Hypothesis (PSSH)

- To apply the *pseudo steady-state hypothesis* assume that the net rate of formation of an active intermediate is zero.

- *The net rate of reaction of an active intermediate, A*, is zero.*

$$r_{A^*} = \sum r_{iA} = 0$$

Determining Reaction Mechanism

1. Assume an active intermediate(s)
2. Postulate a rate law (denominator contains species that collide with active intermediate)
3. Model each mechanism step as an elementary reaction
4. Write the rate laws for 3
5. Assume PSSH
6. Eliminate [A•] terms
7. Check for accuracy vs. experimental data

Nonelementary Kinetics

- What is the rate law for the following reaction?

$$H_2 + Br_2 \rightarrow 2HBr$$

$$r_{HBr} = \frac{k' C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + k'' C_{HBr}}$$

H• and Br•

Rate Order Changes

$$H_2 + Br_2 \rightarrow 2HBr$$

- Consider the hydrogen bromide reaction:
- Which occurs by:
  - Initiation

$$Br_2 \xrightarrow{k_1} 2Br\bullet \quad (1)$$

  - Propagation

$$Br\bullet + H_2 \xrightarrow{k_2} HBr + H\bullet \quad (2)$$

$$H\bullet + Br_2 \xrightarrow{k_3} HBr + Br\bullet \quad (3)$$

$$H\bullet + HBr \xrightarrow{k_4} H_2 + Br\bullet \quad (4)$$

  - Termination

$$2Br\bullet \xrightarrow{k_5} Br_2 \quad (5)$$

Rate Order Changes

- If mechanism steps are elementary, rate can be described as:

(1) $Br_2 \xrightarrow{k_1} 2Br\bullet$

(2) $Br\bullet + H_2 \xrightarrow{k_2} HBr + H\bullet$

(3) $H\bullet + Br_2 \xrightarrow{k_3} HBr + Br\bullet$

(4) $H\bullet + HBr \xrightarrow{k_4} H_2 + Br\bullet$

(5) $2Br\bullet \xrightarrow{k_5} Br_2$

$$r_{HBr} = \sum_{i=1}^{5} r_{iHBr}$$

$$r_{HBr} = r_{2,HBr} + r_{3,HBr} - r_{4,HBr}$$

$$r_{HBr} = k_2 C_{Br\bullet} C_{H_2} + k_3 C_{H\bullet} C_{Br_2} - k_4 C_{H\bullet} C_{HBr}$$

- Need to evaluate $C_{H\bullet}$ and $C_{Br\bullet}$

Rate Order Changes

- Active Intermediate H•:

(1) $Br_2 \xrightarrow{k_1} 2Br\bullet$

(2) $Br\bullet + H_2 \xrightarrow{k_2} HBr + H\bullet$

(3) $H\bullet + Br_2 \xrightarrow{k_3} HBr + Br\bullet$

(4) $H\bullet + HBr \xrightarrow{k_4} H_2 + Br\bullet$

(5) $2Br\bullet \xrightarrow{k_5} Br_2$

$$r_{H\bullet} = \sum_{i=1}^{5} r_{iH\bullet} = 0$$

$$0 = r_{2,H\bullet} - r_{3,H\bullet} - r_{4,H\bullet}$$

$$0 = k_2 C_{Br\bullet} C_{H_2} - k_3 C_{H\bullet} C_{Br_2} - k_4 C_{H\bullet} C_{HBr}$$

$$0 = k_2 C_{Br\bullet} C_{H_2} - C_{H\bullet}\left(k_3 C_{Br_2} + k_4 C_{HBr}\right)$$

$$C_{H\bullet} = \frac{k_2 C_{Br\bullet} C_{H_2}}{k_3 C_{Br_2} + k_4 C_{HBr}}$$

Rate Order Changes

- Active Intermediate Br•:

$$Br_2 \xrightarrow{(Br2)k_1} 2Br\bullet$$

$$Br\bullet + H_2 \xrightarrow{k_2} HBr + H\bullet$$

$$H\bullet + Br_2 \xrightarrow{k_3} HBr + Br\bullet$$

$$H\bullet + HBr \xrightarrow{k_4} H_2 + Br\bullet$$

$$2Br\bullet \xrightarrow{(Br\bullet)k_5} Br_2$$

$$r_{Br\bullet} = \sum_{i=1}^{5} r_{iBr\bullet} = 0$$

$$0 = 2r_{1,Br\bullet} - r_{2,Br\bullet} + r_{3,Br\bullet} + r_{4,Br\bullet} - r_{5,Br\bullet}$$

$$0 = 2k_1 C_{Br_2} - k_5 C_{Br\bullet}^2$$

$$C_{Br\bullet} = \sqrt{\frac{2k_1 C_{Br_2}}{k_5}}$$

known to be zero by PSSH analysis of H•

Rate Order Changes

(1) $Br_2 \xrightarrow{k_1} 2Br\bullet$

(2) $Br\bullet + H_2 \xrightarrow{k_2} HBr + H\bullet$

(3) $H\bullet + Br_2 \xrightarrow{k_3} HBr + Br\bullet$

(4) $H\bullet + HBr \xrightarrow{k_4} H_2 + Br\bullet$

(5) $2Br\bullet \xrightarrow{k_5} Br_2$

- Substituting into r_{HBr}:

$$r_{HBr} = k_2 C_{Br\bullet} C_{H_2} + k_3 C_{H\bullet} C_{Br_2} - k_4 C_{H\bullet} C_{HBr}$$

$$C_{Br\bullet} = \sqrt{\frac{2k_1 C_{Br_2}}{k_5}} \qquad C_{H\bullet} = \frac{k_2 C_{Br\bullet} C_{H_2}}{k_3 C_{Br_2} + k_4 C_{HBr}}$$

$$r_{HBr} = \left(\frac{8k_1 k_2^2}{k_5}\right)^{1/2} \left[\frac{C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + (k_4/k_3) C_{HBr}}\right] = \frac{k' C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + k'' C_{HBr}}$$

Rate Order Changes

- Compare to observation:

$$r_{HBr} = \frac{k' C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + k'' C_{HBr}}$$

$$H_2 + Br_2 \rightarrow 2HBr$$

$$\text{high } C_{Br_2} \rightarrow r_{HBr} \propto C_{Br_2}^{1/2}$$

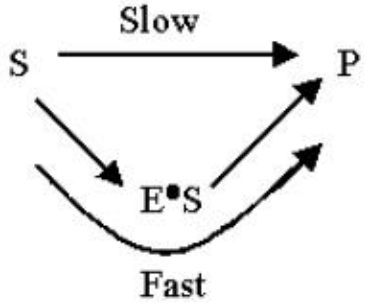
$$\text{low } C_{Br_2} \rightarrow r_{HBr} \propto C_{Br_2}^{3/2}$$

$$\text{high } C_{HBr} \rightarrow r_{HBr} \propto C_{HBr}^{-1}$$

$$\text{low } C_{HBr} \rightarrow r_{HBr} \neq f(C_{HBr})$$

Enzymatic Reaction Fundamentals

- *Enzymes are protein-like substances with catalytic properties.*
- *Enzymes provide a pathway for the substrate to proceed at a faster rate.*
- *The substrate, S, reacts to form a product P*



A given enzyme can only catalyze one reaction. Example, Urea is decomposed by the enzyme urease.

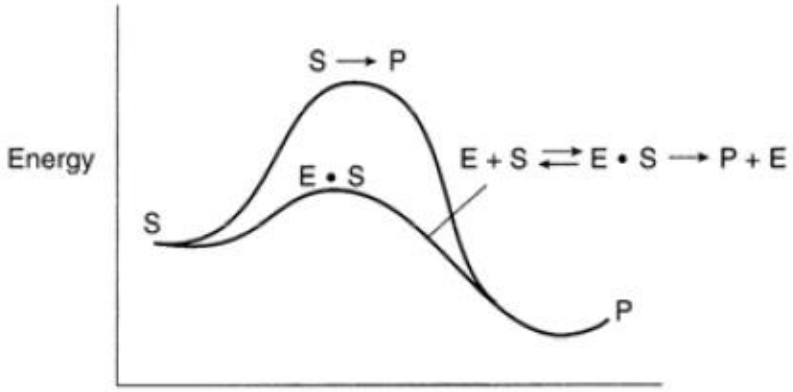
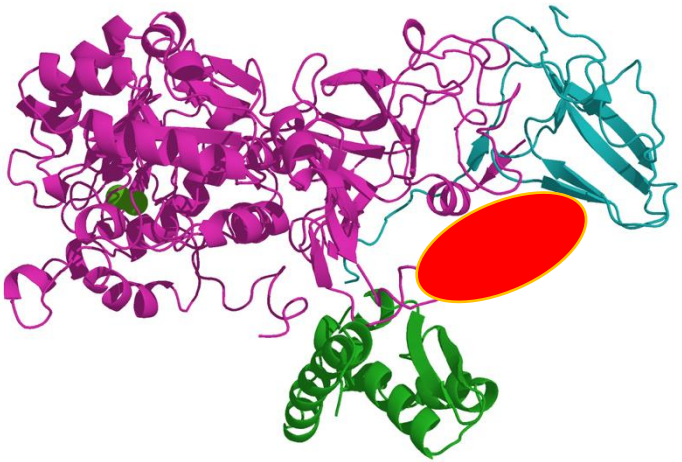


Figure 7-4 Reaction coordinate for enzyme catalysis.

Enzymes

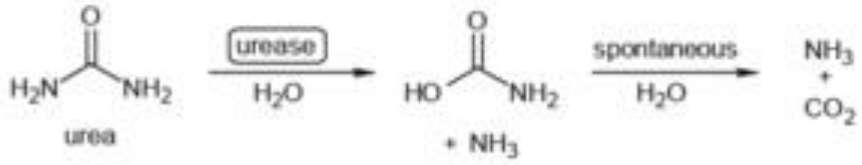
Michaelis-Menten Kinetics

Enzymes are protein-like substances with catalytic properties.

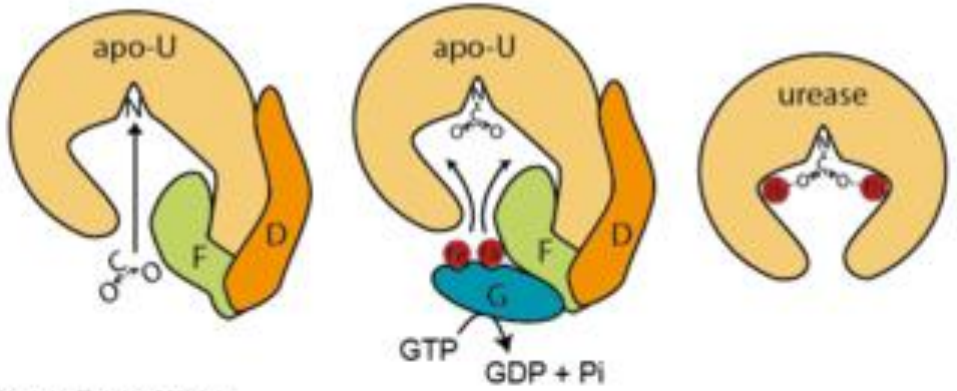


Enzyme Urease

urease reaction



model of urease activation



Claus-Peter Witte
FU Berlin, 2010

Enzymes - Urease

A given enzyme can only catalyze only one reaction. Urea is decomposed by the enzyme urease, as shown below.

$$NH_2CONH_2 + UREASE \xrightarrow{H_2O} 2NH_3 + CO_2 + UREASE$$

$$S + E \xrightarrow{H_2O} P + E$$

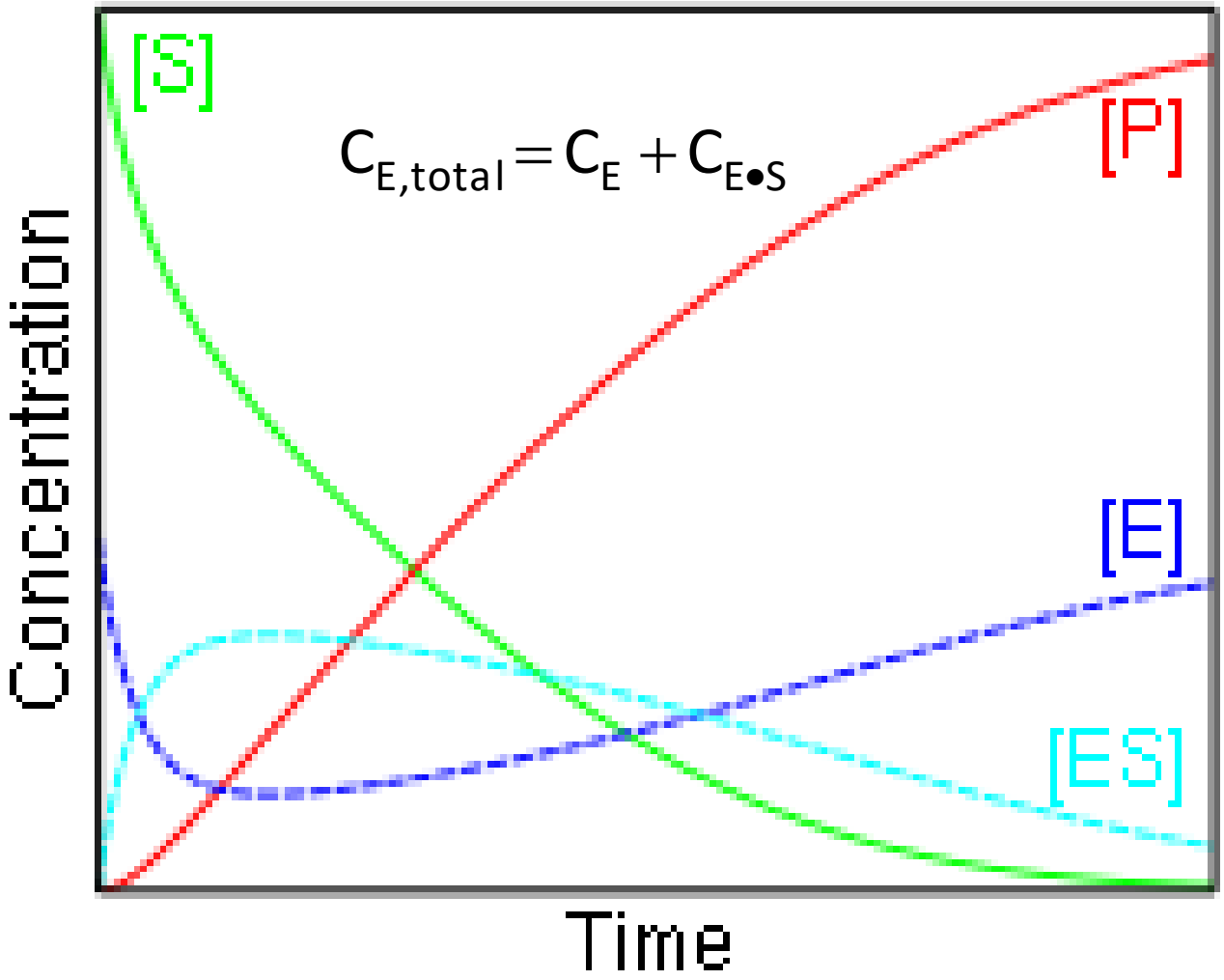
The corresponding mechanism is:

$$E + S \xrightarrow{k_1} E \bullet S$$

$$E \bullet S \xrightarrow{k_2} E + S$$

$$E \bullet S + W \xrightarrow{k_3} P + E$$

Michaelis-Menten Kinetics



Michaelis-Menten Kinetics

- Model for substrate/enzyme interactions

$$E + S \xrightarrow{k_1} E \bullet S$$

$$E \bullet S \xrightarrow{k_2} E + S$$

$$E \bullet S + W \xrightarrow{k_3} P + E$$

E ≡ enzyme (catalyst)
S ≡ substrate (chemical reacted)
W ≡ water (always in high excess)

$$r_{E\bullet S} = 0$$
$$= k_1 C_E C_S - k_2 C_{E\bullet S} - k_3 C_{E\bullet S} C_W$$
$$= k_1 \left(C_{E,total} - C_{E\bullet S} \right) C_S$$
$$- k_2 C_{E\bullet S} - k_3 C_{E\bullet S} C_W$$

$$C_{E,total} = C_E + C_{E\bullet S}$$

$$r_P = k_3 C_{E\bullet S} C_W$$

$$C_{E\bullet S} = \frac{k_1 C_{E,total} C_S}{k_1 C_S + k_2 + k_3 C_W}$$

Michaelis-Menten Kinetics

- Model for substrate/enzyme interactions

$$E + S \xrightarrow{k_1} E \bullet S$$

$$E \bullet S \xrightarrow{k_2} E + S$$

$$E \bullet S + W \xrightarrow{k_3} P + E$$

E ≡ enzyme (catalyst)
S ≡ substrate (chemical reacted)
W ≡ water (always in high excess)

$$r_P = \frac{k_1 k_3 C_{E,total} C_W C_S}{k_1 C_S + k_2 + k_3 C_W}$$

$$C_{E,total} = C_E + C_{E \bullet S}$$

$$r_P = k_3 C_{E \bullet S} C_W$$

$$C_{E \bullet S} = \frac{k_1 C_{E,total} C_S}{k_1 C_S + k_2 + k_3 C_W}$$

Michaelis-Menten Kinetics

- Model for substrate/enzyme interactions

$$E + S \xrightarrow{k_1} E \bullet S$$

$$E \bullet S \xrightarrow{k_2} E + S$$

$$E \bullet S + W \xrightarrow{k_3} P + E$$

E ≡ enzyme (catalyst)
S ≡ substrate (chemical reacted)
W ≡ water (always in high excess)

$$r_P = k_3 C_{E \bullet S} C_W$$

$$r_P = \frac{k_1 k_3 C_{E,total} C_W C_S}{k_1 C_S + k_2 + k_3 C_W}$$

$$k_3' = k_3 C_W$$

Turnover number

$$K_M = \frac{k_3' + k_2}{k_1}$$

K_M ≡ Michaelis Constant (affinity constant)

$$-r_S = \frac{k_3' C_{E,total} C_S}{C_S + K_M}$$

Michaelis-Menten Kinetics

- Model for substrate/enzyme interactions

$$E + S \xrightarrow{k_1} E \bullet S$$

$$E \bullet S \xrightarrow{k_2} E + S$$

$$E \bullet S + W \xrightarrow{k_3} P + E$$

E ≡ enzyme (catalyst)
S ≡ substrate (chemical reacted)
W ≡ water (always in high excess)

$$r_P = k_3 C_{E \bullet S} C_W$$

$$r_P = \frac{k_1 k_3 C_{E,total} C_W C_S}{k_1 C_S + k_2 + k_3 C_W}$$

Maximum rate for a given enzyme (V_{max})

$$-r_S = \frac{V_{max} C_S}{C_S + K_M}$$

$$V_{max} = k_{cat} E_t$$

$$k_3' = k_3 C_W$$

Turnover number

$$K_M = \frac{k_3' + k_2}{k_1}$$

K_M ≡ Michaelis Constant (affinity constant)

$$-r_S = \frac{k_3' C_{E,total} C_S}{C_S + K_M}$$

Michaelis-Menten Kinetics

$$V_{max} = k_{cat} E_t$$

Turnover Number: k_{cat}

Number of substrate molecules (moles) converted to product in a given time (s) on a single enzyme molecule *(molecules/molecule/time)*

For the reaction: $H_2O_2 + E \xrightarrow{k_{cat}} H_2O + O + E$

? molecules of H_2O_2 converted to product per second on a single enzyme molecule.

Michaelis-Menten Kinetics

$$V_{max} = k_{cat} E_t$$

Turnover Number: k_{cat}

Number of substrate molecules (moles) converted to product in a given time (s) on a single enzyme molecule *(molecules/molecule/time)*

For the reaction: $H_2O_2 + E \xrightarrow{k_{cat}} H_2O + O + E$

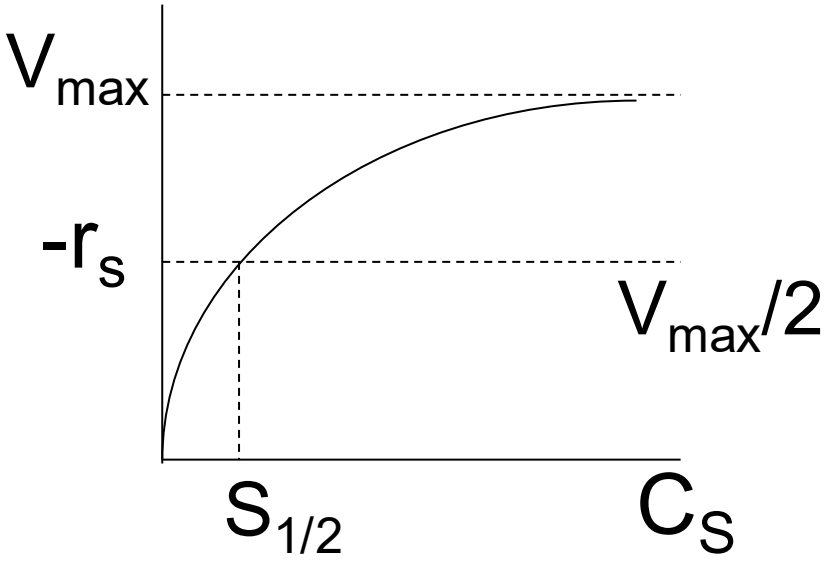
40,000,000 molecules of H_2O_2 converted to product per second on a single enzyme molecule.

Michaelis-Menten Kinetics

Michaelis-Menten Equation

$$r_P = -r_S = \frac{V_{max} S}{K_M + S}$$

(Michaelis-Menten plot)



Solving: $\frac{V_{max}}{2} = \frac{V_{max} S_{1/2}}{K_M + S_{1/2}}$

$K_M = S_{1/2}$

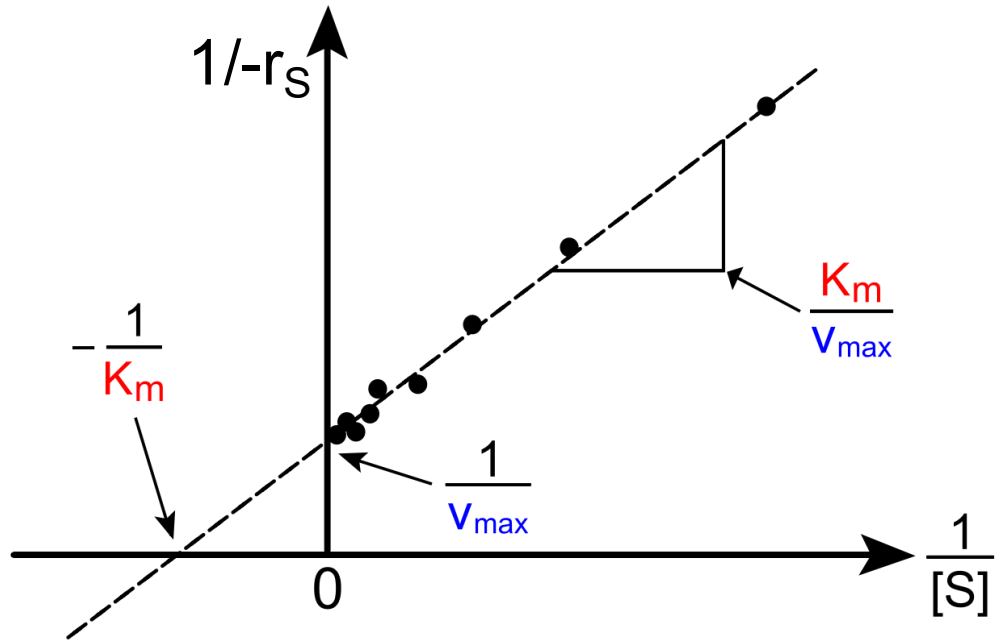
Therefore, K_M is the concentration at which the rate is half the maximum rate.

Michaelis-Menten Kinetics

Inverting yields:

$$\frac{1}{-r_S} = \frac{1}{V_{max}} + \frac{K_M}{V_{max}}\left(\frac{1}{S}\right)$$

Lineweaver-Burk Plot



See:
Page 401-403
Page 404-406
Example 7.3

Types of Enzyme Inhibition

(P409)

1. Competitive

$$E + I \Leftrightarrow I \bullet E \,(\text{inactive})$$

2. Uncompetitive

$$E \bullet S + I \Leftrightarrow I \bullet E \bullet S \,(\text{inactive})$$

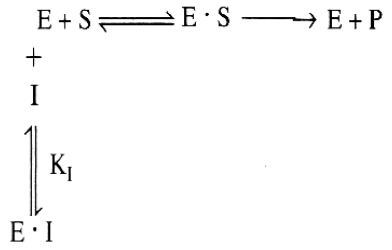
3. Non-competitive

$$E \bullet S + I \Leftrightarrow I \bullet E \bullet S \,(\text{inactive})$$

$$I \bullet E + S \Leftrightarrow I \bullet E \bullet S \,(\text{inactive})$$

1. Competitive Inhibition:

Competitive inhibition pathway



Reaction Steps

(1) $E + S \xrightarrow{k_1} E \bullet S$

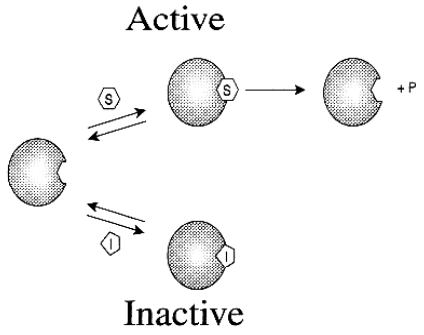
(2) $E \bullet S \xrightarrow{k_2} E + S$

(3) $E \bullet S \xrightarrow{k_3} P + E$

(4) $I + E \xrightarrow{k_4} E \bullet I$ (inactive)

(5) $E \bullet I \xrightarrow{k_5} E + I$

Competitive Inhibition Pathway



(a) Competitive inhibition. Courtesy of D. L. Nelson and M. M. Cox, *Lehninger Principles of Biochemistry*, 3rd ed. (New York: Worth Publishers, 2000), p. 266.

Competitive Inhibition:

$$E + S \underset{k_2}{\overset{k_1}{\rightleftharpoons}} E \bullet S \xrightarrow{k_3} E + P$$

$$E + I \underset{k_5}{\overset{k_4}{\rightleftharpoons}} E \bullet I \text{ (inactive)}$$

1) Mechanisms:

(1) $E + S \xrightarrow{k_1} E \bullet S$

(2) $E \bullet S \xrightarrow{k_2} E + S$

(3) $E \bullet S \xrightarrow{k_3} P + E$

(4) $I + E \xrightarrow{k_4} E \bullet I$ (inactive)

(5) $E \bullet I \xrightarrow{k_5} E + I$

Competitive Inhibition:

(1) $E + S \xrightarrow{k_1} E \bullet S$

(2) $E \bullet S \xrightarrow{k_2} E + S$

(3) $E \bullet S \xrightarrow{k_3} P + E$

(4) $I + E \xrightarrow{k_4} E \bullet I$ (inactive)

(5) $E \bullet I \xrightarrow{k_5} E + I$

2) Rate Laws:

$$r_{E\cdot S} = 0 = k_1 C_S C_E - k_2 C_{E\cdot S} - k_3 C_{E\cdot S}$$

$$C_{E\cdot S} = \frac{k_1 C_S C_E}{k_2 + k_3} = \frac{C_S C_E}{K_m}$$

$$r_P = \frac{k_3 C_S C_E}{K_m}$$

$$r_{I\cdot E} = 0 = k_4 C_I C_E - k_5 C_{I\cdot E}$$

$$C_{I\cdot E} = \frac{C_I C_E}{K_I} \qquad K_I = \frac{k_5}{k_4}$$

Competitive Inhibition:

$$C_{I\cdot E} = \frac{C_I C_E}{K_I} \qquad C_{E\cdot S} = \frac{C_S C_E}{K_m}$$

$$C_{Etot} = C_E + C_{E\cdot S} + C_{I\cdot E}$$

$$C_E = \frac{C_{Etot}}{1 + \frac{C_S}{K_m} + \frac{C_I}{K_I}}$$

$$r_P = \frac{k_3 C_{Etot} C_S}{K_m + C_S + \frac{C_I K_m}{K_I}}$$

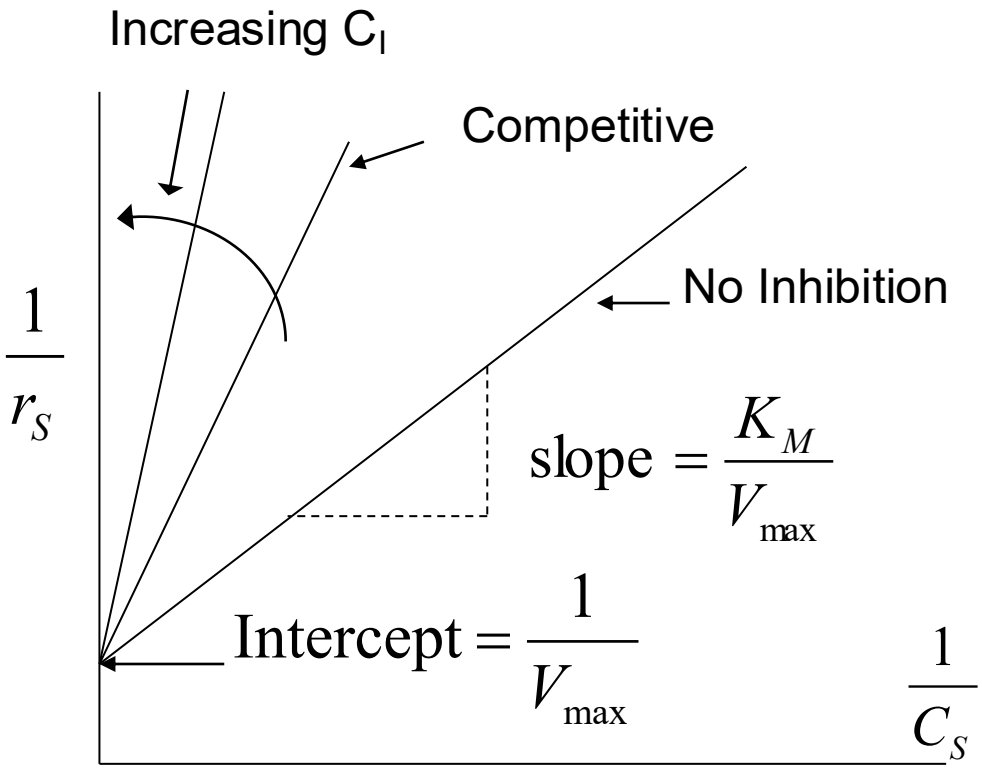
$$-r_S = \frac{V_{max} C_S}{C_S + K_m\left(1 + \frac{C_I}{K_I}\right)}$$

$$\frac{1}{-r_S} = \frac{1}{V_{max}} + \frac{k_m}{V_{max}}\left(1 + \frac{C_I}{K_I}\right)\frac{1}{C_S}$$

Competitive Inhibition:

From before (no competition): $$\frac{1}{-r_S} = \frac{1}{V_{max}} + \frac{K_M}{V_{max}}\frac{1}{C_S}$$

Lineweaver-Burk Plot



Competitive

$$\frac{1}{-r_S} = \frac{1}{V_{max}} + \frac{K_M}{V_{max}}\left(1 + \frac{C_I}{K_I}\right)\frac{1}{C_S}$$

Intercept does not change, slope increases as inhibitor concentration increases.